EVOLVE OR STEAL

GAME DESIGN DOCUMENT | VERSION 0.4

Final currency architecture, monetization strategy, pet incubation and experimental balance design

Prepared: 15 October 2026 | Roblox product/design handoff | Editable design specification

Executive verdict. *Use an earned Gold economy for frequent, mostly temporary expression; a purchasable + earnable Gem economy for high-value permanent cosmetics; and play-earned egg materials for pet acquisition. Gem-paid focusing is an explicitly gated experiment, not a default launch entitlement.*

Design status at a glance

System	Decision	State
Three-round infection chase	Runner to Catcher conversion, no spectating	LOCKED
Arena egg theft	Real transfer of current egg; highest material reward	LOCKED
Starting pet	None; first Common earned through matches	LOCKED
Pets and abilities	One equipped pet grants one family ability	LOCKED
Focused incubation	Extra same-tier egg piece; first Common focus free	LOCKED
Gold	Earn-only, frequent low-cost expression	RECOMMENDED
Gems	Earnable AND Robux-purchasable	LOCKED
Gem-powered focusing	Optional alternative to extra material	EXPERIMENT / POLICY GATE
Casual rarity stats	0 vs +1.5/+3/+4.5% test	EXPERIMENT
Ranked	Normalize pet rarity stats; launch only if queue supports it	DEFER

Important distinction: This document is a proposed business/game economy based on public Roblox documentation, independent Huss Valley reporting and mathematical scenarios. No private competitor revenues or true player conversion rates were available.

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Status vocabulary: LOCKED = selected for prototype; RECOMMENDED = default configuration needing tests; EXPERIMENT = instrumented A/B alternative; DEFER = not required at launch.

01 Evidence and decision thesis

Core product thesis. *Monetization matters to Roblox discovery, but retention and low bounce carry more weight. Build multiple attractive purchases without selling an overwhelming advantage or compromising the chase.*

- Roblox lists play-through, early bounce, play days and playtime as its most important recommendation factors; spend days and Robux spent per user also matter. Organic Home-acquired users drive ranking signals. [S1][S2]
- Roblox advises separating repeatable consumables from durable purchases, offering different price points, and tracking currency sources/sinks and the acquisition source of players. [S3][S4][S5]
- Huss Valley appears to use Gems for permanent abilities and Coins for one-match consumables. Our ability progression is already tied to PETS, so duplicating an independent Gem ability shop would hollow out the egg loop. [S6][S7]
- Psychology: autonomy and competence support engagement; visible goals support persistence in certain contexts; competitive purchases can cause fairness concerns. None supplies an optimal currency price or conversion rate. [S10][S11][S12]

02 Gameplay and roles

Phase	Runner	Catcher / Hunter
Match start	One temporary Level-1 arena egg; no starter pet required.	One starter Catcher volunteers, fair fallback if none; no egg required.

Crossing 1	Reach safe side, egg transforms to Level 2.	Catch Runners; victim converts at phase transition.
Crossing 2	Egg transforms to Level 3 if survived.	Steal eggs and grow Catcher team.
Crossing 3	Secure Level 4 if survived; all levels reset next match.	Surround remaining Runners; last-chance steals.
Match end	Earn best qualifying egg material plus Gold/Gems.	Earn highest stolen-material tier plus Gold/Gems.

Default 8-16 players; target 135-second match cycle including intermission; hard cap 165 seconds. Three crossings. Infection is one-way during a match. No player is eliminated into spectators. Server finalizes each capture atomically and preserves ability cooldown on role conversion.

A Catcher visually steals a physical egg at capture time, but the egg enters a non-physical per-match tally; Catchers do not accumulate floating egg stacks. Each player gets at most one PRIMARY egg material after a match, determined by their best eligible result. Previously completed crossings earn a banked minimum Common-equivalent reward; the victim keeps Gold/Gems already accrued.

03 Egg materials and incubation

Highest level attained	One material tier	Craft default	Focused (same tier)
Level 1 / early capture	Common	2 pieces	3 pieces
Level 2	Rare	3 pieces	4 pieces
Level 3	Epic	5 pieces	6 pieces
Level 4 / secure final	Epic piece + victory bonus at launch	5 Epic pieces	6 Epic pieces

Common exceptions: the FIRST EARNED Common pet requires 2 materials and includes a FREE chosen ability family. There is no pet on initial spawn. Afterward, players may select random family for base cost or selected family for one extra piece. Fixed rarity comes from the material tier; do NOT add a separate random rarity roll. One ability family corresponds to one collectible species per tier initially; no additional species RNG in paid-influenced flows.

Every completed active match grants at least a Common-equivalent piece. Higher-tier materials may be voluntarily downgraded 1:1 to fill lower-tier recipes, never upgraded upward. This ensures a first Common pet can be earned after two active matches even if a player won higher-tier materials.

Higher-rarity materials are earned from conditional outcomes, not necessarily awarded by every winner. IMPORTANT: Crowned/Legendary pieces are DEFERRED at launch; a Level-4 winning egg currently secures an Epic piece plus the separate win Gold/Gem bonus. A later live season may add a Crowned piece route only after tuning. Never label a reward as Legendary while granting unrelated material.

Primary material reward is capped at one per match. The material cap does not stop Coins/Gems, quests or non-item recognition from rewarding multiple catches.

04 Why Gold exists

Gold mission. *A high-frequency earn-only soft currency for low-stakes purchases and recurring expression. It should not buy abilities, eggs, rarity upgrades or Gem equivalents.*

Sink	Gold price	Duration	Reason
Egg trail / shell aura for one match	35-60	Consumable	Repeatable expressive sink without capture advantage
Lobby confetti burst / victory flare	75	Consumable	Co-play and spectator expression, not combat

Basic emote	500	Permanent	First affordable purchase around 27 min typical active play
Carry pose / egg case skin	900	Permanent	Meaningful medium-term identity goal
Lobby entrance animation	1,400	Permanent	Aspirational free progression
Weekly Gold colorway	2,400	Permanent, rotating	Longer Gold goals; clearly published stock and price

Gold cannot be purchased with Robux or exchanged for Gems at launch. No Gold-purchasable random crate will be indirectly funded by purchased currency. The repeatable purchases are deliberately COSMETIC, not Huss Valley-style gameplay-affecting consumables, because pets already occupy the ability slot.

Gold design failure: If players buy permanent cosmetics only once, balances compound and the soft currency becomes worthless. The store therefore requires content refreshes and opt-in consumable uses; never wipe balances or penalize saving. Track balance growth by cohort rather than forcing consumption.

05 Why Gems exist

Gem mission. *A premium but earnable currency for high-value durable expression. Selling Gems with Robux is part of the proposed launch. It does not automatically entitle a player to an egg or pet.*

Sink	Gems	Purchase type	Economy rationale
Premium egg trail theme	100-180	Permanent, guaranteed	Small first meaningful purchase
Pet ability VFX color pack	250	Permanent, guaranteed	Shows off chosen family; same hitboxes and duration
Hatching ceremony / animation	350	Permanent, guaranteed	Makes earned hatch feel special
Pet showcase / incubator display slot	450	Permanent, guaranteed	Ownership identity between rounds; no combat slots
Legendary-themed visual override	700	Permanent, guaranteed	High-value cosmetic tier, requires family already owned
Prestige presentation set	1,500	Permanent, guaranteed	Long-term cosmetic goal; no combat stats
Focus Seal by tier	4 / 9 / 16 / 60	EXPERIMENT ONLY	Replaces extra focused-hatch piece, never base requirement

The Gem Focus Seal is NOT a launch-default feature. It may be introduced only following paid-random-item/policy review, a deterministic outcome implementation, and evidence that paid players are not materially outperforming unpaid players. If the purchased Gem can influence a random outcome, Roblox treats the system as paid-random or probability-modifying; numerical outcome disclosure and player eligibility enforcement apply. [S8]

If Focus Seals are enabled, they replace ONLY the incremental material used for family focus. They never replace the core 2/3/5 pieces required to hatch. Do not sell egg materials, a higher starting egg level, superior cooldowns, or Legendary rarity rolls through Gems at launch.

06 Robux monetization catalog

Developer product: purchasable Gems	Robux	Gems / Robux
100 Gems	49	2.04
225 Gems	99	2.27
500 Gems	199	2.51
1,100 Gems	399	2.76
2,400 Gems	799	3.00
5,000 Gems	1499	3.34

Catalog strategy. *Offer a small entry price, two mid-market Gem packs, and larger options for collectors; monitor payer conversion and purchase satisfaction. Pack prices are candidate prices, not benchmark-optimal results.*

Other product	Candidate price	Type	Entitlement
Custom lobby celebration	79	Pass	Permanent, fixed effect
2x Gold	199	Pass	Doubles cosmetic-only soft currency; no Egg/Gem bonus
Pet FX Bundle	249	Pass	Guaranteed family visual pack, not the pet itself
VIP	499	Pass	Nameplate, collection display style, one premium emote
Premium collection track	799	Pass, DEFER	Guaranteed visual cosmetics and display items; free track also exists
Prestige pet visual skin	1299	Pass/product, DEFER	Skin only; must own corresponding pet family

Developer product purchases: use server receipt processing and durable, idempotent transaction grants. Persist paid Gem credit separately by transaction ID and merge into spendable wallet only after the receipt succeeds; log acquired-from-Robux tags for auditing. Never show misleading sales countdowns, fake discounts, or interrupt an active capture with a purchase modal. [S9][S3]

Roblox generally pays developers ~70% of ordinary in-experience Robux sales; eligibility and DevEx rates vary. Pricing and sale details can change and must be verified at Creator Dashboard before launch. [S13][S14]

07 Huss Valley competitor audit and what transfers

Huss Valley observed / reported	Confidence	Evolve or Steal response
Gems for permanent abilities; Coins for single-use items	Secondary guides agree	Use premium Gems for permanent expressive goods; Gold for affordable/repeatable expression
XP boosts and VIP: 2x EXP 199; VIP 799	Public Rolimon snapshots	Offer Gold convenience and VIP prestige; avoid egg/power multipliers by default
Premium 799; high-tier knife cosmetics 1,799	Public Rolimon snapshots	High-ticket guaranteed visual pet skins considered AFTER demand validation
Crates 750 Gems; odds 1.5%-30% / item	Independent in-game screen report	Avoid importing extreme rarity RNG into competitive ability access
Many developer products listed in third-party trackers	Incomplete public audit	Do not assume private revenue distribution or all current product IDs

Huss Valley is a useful ladder model, but its ability sales do not map 1:1 to our game because pets, not Gems, must grant abilities. Public listings indicate catalog breadth, not sales conversion, margins, payer composition, or customer satisfaction. [S6][S7][S15]

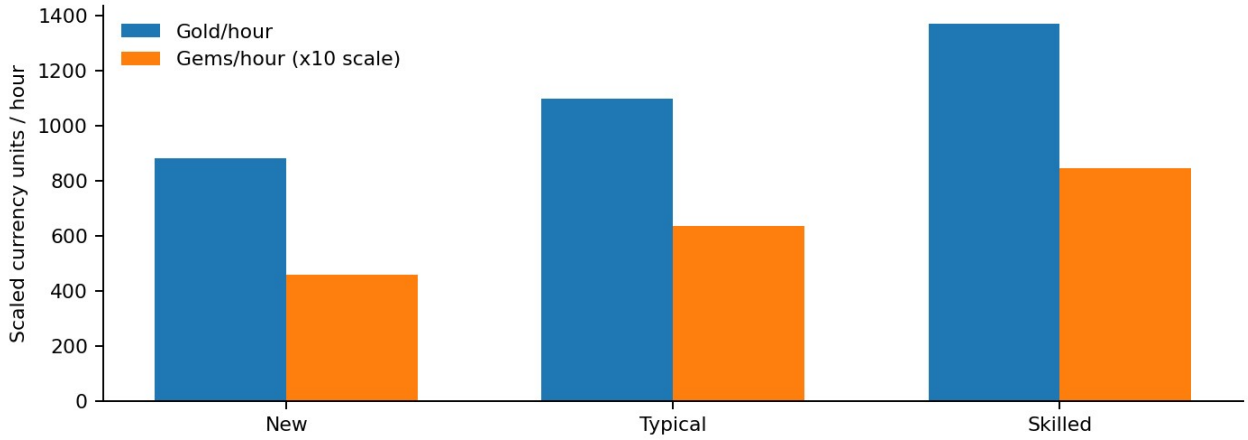
08 Reward emission model (scenario inputs)

This STRESS TEST models the prospective full four-tier economy, INCLUDING a Crowned tier that is DEFERRED FROM LAUNCH. New, Typical and Skilled player cohorts receive one primary piece per eligible match based on best qualifying escape/theft outcome. To avoid silently assuming all catches are successes, the scenario collapses each match into a categorical material outcome. These probabilities are hypothetical; they must be replaced by actual gameplay telemetry. At launch, Crowned outcomes roll down into Epic until the fourth tier is enabled.

Material outcome / match	New	Typical	Skilled
Common	66%	40%	20%
Rare	27%	36%	30%
Epic	6.5%	20%	38%
Crowned	0.5%	4%	12%

Gold earned = min(90, 25 active + 12 x successful crossings + 10 x credited first theft + 15 x final victory). Gems earned = min(5, 1 active +1 first crossing +1 second crossing +1 final victory +1 credited first theft). The model approximates credited-theft probabilities of 30%, 50%, and 75% across New/Typical/Skilled, respectively. Reward components are earned ONLY after server-confirmed contributions, never by AFK presence.

Scenario output	New	Typical	Skilled
Gold per match	33.0	41.2	51.4
Earned Gems per match	1.71	2.38	3.17
Gold/hour	881	1099	1369
Gems/hour	46	64	85



Scenario values use simulated averages over 150,000 match outcomes/cohort. For chart readability, Gems are scaled by 10 and should NOT be interpreted as directly equivalent to Gold.

09 Acquisition math and focusing fairness

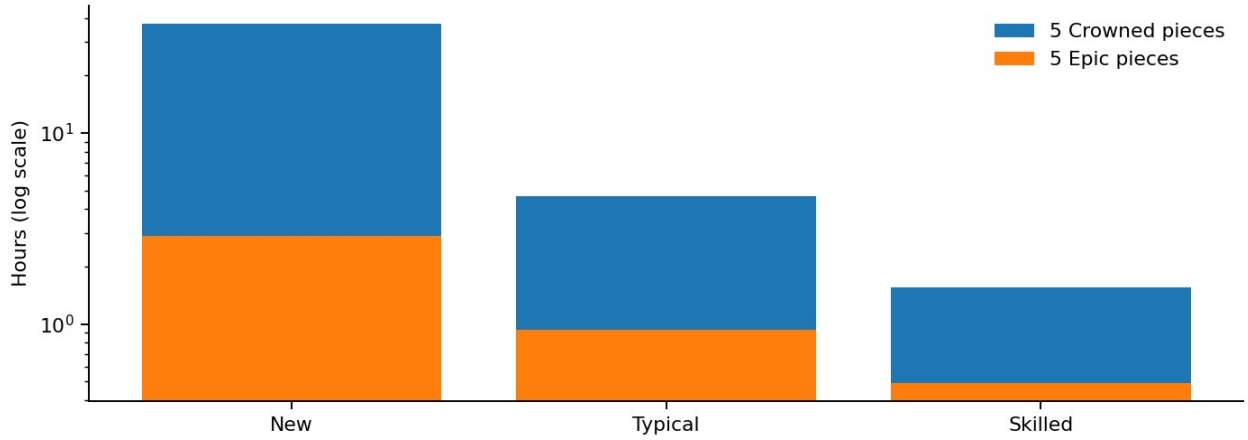
For K=9 equipable ability families, random family selection has P(target family)=1/9. One random hatch costs 3 materials under the standard Rare recipe; expected random cost to get one targeted family = 3/(1/9)=27 materials. A focused hatch costs 4 and guarantees the family. The first Common hatch is exception: 2 pieces with free family choice.

Expected materials for all 9 different family abilities: random only = 3*K*H_K = 76.38; focus only = 4*K = 36; optimal switch from random when >=7 families are missing to focus when <=6 remain = 34.23 materials. This uses independent uniform family probabilities and the 3/4 recipe, NOT a guarantee of real collection time.

A 50% focused share reduces unit-output per egg material by 1 - 3/3.5 = 14.3% versus all-random hatching. Focusing is a player-agency mechanism, NOT the solution to egg oversupply.

Anti-frustration rule. *First Common pet always grants chosen ability family on the base recipe; subsequent families can be obtained deterministically via extra-piece Focus. This retains meaningful pets without creating RNG-only access to critical playstyles.*

10 Rare material stress test and protection



Simple expected time to acquire five exact-tier pieces: E[time] = (5/p) x 2.25 min. Log scale highlights severe high-tier inequality under hypothetical drop rates. Acquiring a different tier does not count unless a clearly defined conversion is used.

Five required pieces	New	Typical	Skilled
Epic expected hours	2.88	0.94	0.49
Crowned expected hours	37.5	4.69	1.56

The new-versus-skilled Crowned gap is 24x (37.5h versus 1.56h under the modeled probabilities). This is NOT acceptable as a launch progression model if Crowned also confers mechanically better casual abilities. The GDD therefore marks Crowned rarity as a deferred endgame tier until guaranteed paths and actual acquisition distributions are established.

Candidate protection: a deterministic Crowned material award after 35 meaningfully played matches, stacking with natural rewards and with a per-account track that cannot be AFK-farmed. Even if natural Crowned drops never occur, five such milestones take at most 175 matches, or 6.56 active hours. This is an illustrative cap and may be too generous for future endgame content. Do not promise it before testing.

The alternative is to launch with Common/Rare/Epic only and introduce Crowned later with non-RNG achievement routes. This is the preferred commercialization decision. A simple sink or bigger copy requirement does not fix a 24x skill acquisition gap.

11 Gold/Gem sources versus sinks

At the Typical modeled rate: about 1,099 Gold/hour and 64 Gems/hour. A 500-Gold emote takes ~27 minutes of uninterrupted active play; a 1,400-Gold presentation animation takes ~76 minutes. A 150-Gem premium trail requires ~2.36 earned-play hours; a 450-Gem showroom feature requires ~7.08 hours. Paid Gem packs shorten cosmetic acquisition, not the core egg materials.

Typical-player spending scenario	Source / match	Sink / match	Result
Gold: cautious adoption	41.2	(15% x 50) + (600/40) = 22.5	18.7 Gold/match net accumulation
Gold: active adopters	41.2	(30% x 50) + (600/25) = 39.0	2.2 Gold/match net accumulation
Gold: heavy cosmetic spending	41.2	(50% x 50) + (600/20) = 55.0	Spends more than newly earned Gold; wallet drawdown
Gems: cosmetic-only reserve	2.38 earned	Sinks are discretionary	Balance grows when collection content stalls

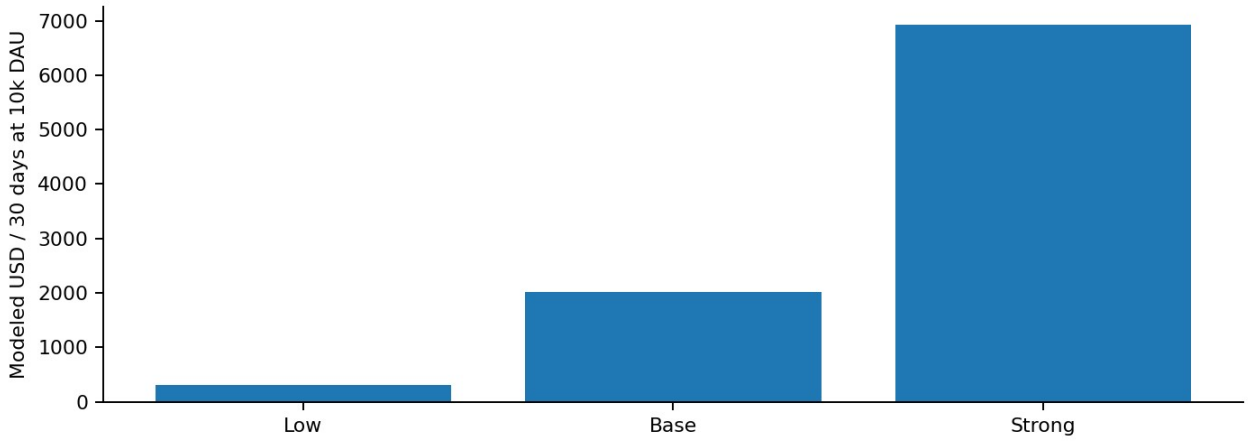
Interpretation: there is no guaranteed stable sink equilibrium. The studio must update a rotating Gold catalog, review average wallets by playtime, and create durable premium Gem products players genuinely want. Avoid forcing spend with artificial expiry or hidden upkeep.

Economy dashboard checks weekly: earned Gold, Gold spent, average Gold wallet by new/returning users, purchased Gems, earned Gems, Gem expenditures by item, unspent purchased-Gem balance, hatch materials by rarity, expired purchases, shop conversion, spend days, and refund rates.

12 Revenue sensitivity & discovery reality

Cash-out scenario: 10,000 average DAU for 30 days, each day a fixed fraction of DAU makes a purchase of a specified mean price. Revenue applies a simple 70% creator share and standard $0.0038 DevEx rate per Earned Robux. Does NOT include creator rewards, refunds, taxes, regional economics, user lifetime, server costs or acquisition spending. Exact accounts may vary.

Scenario	Daily payer rate	Mean Robux / purchase	Creator USD / 30d
Low	0.5%	79	$315
Base	1.5%	169	$2,023
Strong	3.0%	289	$6,919



Constructed earnings scenarios for 10,000 average DAU; these are not Huss Valley numbers or business forecasts.

Algorithm reality. *Home ranking explicitly includes spend days and Robux spent per user, but retention/bounce/play days/playtime are priority signals. Only behavior of users organically acquired from Home Recommendations drives its ranking metrics. Spend more intelligently, not more intrusively. [S1][S2]*

Monetization hypothesis: 49-99 Robux premium Gem packs improve accessible payer conversion; 199-399 packs help engaged users; 799+ packs target high-interest cosmetic collectors. We cannot know optimal prices without purchase data. Roblox-managed price optimization ordinarily requires high transaction volume. [S16]

13 Pet abilities, tiers and visibility

One equipped pet means one equipped gameplay ability; NO starter pet. The pet is NOT a purely cosmetic item. Nine planned family archetypes: Dash, Decoy, Scout, Brake, Spring, Mist, Rebound, Drift, Tracker. Launch with three polished abilities, expand to six after tests, and reach nine only after observing fair use across both roles.

Pet rarities within a family share ability ID. Casual mechanical tiers are TEST-ONLY: Common +0%, Rare +1.5%, Epic +3%, future Crowned +4.5%, improving only one family-specific parameter (distance OR duration OR recharge). Effects do not stack. Ranked, if later enabled, normalizes all rarity parameters. Note that even small cooldown reductions can change possible activation counts at fixed match boundaries.

Visibility: owner sees one small cosmetic follower rendered locally; opponents do not see persistent follower models but do see a short pet summon and gameplay telegraph when an ability fires. Lobby, collection, and victory podium allow pet display subject to distance/device limits. Server owns ability effects and cooldown; clients own cosmetic follower models only.

No pet player starts with the same baseline movement and dodge system. Test that new players retain a meaningful chance to escape/capture before their first hatch. If no-pet disadvantage is too high, reduce ability effects or add standardized baseline counters rather than gifting pets.

14 FTUE / purchase timing / Ranked

1. Spawn and immediately enter a legible match; introduce egg evolution and catching, not currency shopping.
2. By 1 eligible match, show one earned material and progress to the first pet (1 of 2).
3. By 2 meaningful matches, let player craft first Common pet from earned pieces with free family selection; do not make them pay extra for the first useful ability.
4. Show Gold as a useful optional expression currency and Gems as a premium optional convenience/visual currency only after the player experiences earning/catching eggs.
5. Offer paid Gem packs in the persistent shop and the incubator lobby, not during active chases. Paid focuses should remain disabled until legal/policy and fairness conditions pass.
6. Ranked requires adequate queue population; future Ranked standardizes pet rarity stats but must also ensure practical access to all ability families or provide standardized loaners.

UX: hide economy counters during critical crossing except one unobtrusive egg level indicator and ability cooldown. Show one end-of-match consolidated breakdown: Gold, Gems, best stolen/secured egg material, recipe progress, collection milestone. All purchases display exact delivered currency/entitlement and actual final price.

15 Technology and consumer protections

Subsystem	Server requirements
Capture/role change	Server validation of hitbox, latency window, cooldown; atomic role+egg transfer
Wallets	Gold earned balance; Gem earned and Gem purchased ledger by transaction ID; primary material stack by rarity
Store	Robux pricing resolved from MarketplaceService; ProcessReceipt grants paid Gems exactly once
Crafting	Atomic debit of required pieces, optional Focus Seal, unique pet record; rollback on failure
Policies	Use PolicyService per player for any purchasable-currency-influenced random item; restrict feature as required

Anti-farm	Server-side time/activity rules; credited steal constraints; no self-capture or coordinated farming loops
Analytics	Log economy SOURCE/SINK after completion, funnel conversions, cohort role performance

If purchased Gems influence ANY randomized hatch outcome directly or indirectly, Roblox requires actual item-level numerical odds and restrictions for disallowed players; items that change RNG probabilities are expressly covered. A guarantee of chosen family is not sufficient to dismiss policy concerns if other random components remain. [S8]

For direct deterministic Gem purchases, show outcome and charge only when the user confirms. Avoid dark-pattern urgency. Separate subscriptions and bundles require explicit return obligations and item descriptions. [S3][S9]

16 Analytics and experimental design

Hypothesis	Primary test	Stop/hold trigger
Gold is useful	Gold sink/source, ownership, shop-open-to-purchase	Most repeat visitors accumulate Gold without spending after a content rotation
Gems are attractive without P2W	Gem pack payer rate, purchase satisfaction, D1/D7 by paid vs nonpaid	Early player departures or strong paid win disparity
First pet timing supports onboarding	Two-match pet unlock funnel	Large % fail to reach first pet despite active play
Focus reduces ability RNG frustration	Random vs Focus selection; family unlock time	Players forced to random-hatch for main playstyle
Rarity stats acceptable	Skill-adjusted escape/capture win differences; mobile cohorts	Rarity dominates actual skill outcomes
Premium sale serves RFY	Home organic cohort spend-day and bounce trend	Revenue rises but bounce/retention worsens
Casual is fast and fair	Match time, post-capture engagement, zero-runner outcomes	Idle Chasers, long waits, capture snowball too fast

Recommended experiment sequence: P0 playtest gameplay without offers; P1 currencies and Gold shop; P2 purchasable Gems with strictly guaranteed cosmetics; P3 test Gem Focus Seal only if policies and fairness pass; P4 investigate Casual rarity stats with controlled skill-matched cohorts; P5 queue-limited Ranked if enough users.

A/B trials require enough independent organic players for meaningful conclusions. A 20-40-person playtest diagnoses usability failures; it cannot reliably identify a small retention gain or conversion lift.

17 Implementation roadmap and launch scope

Milestone	Deliverable	Release condition
Week 1 / P0	Three-crossing chase, eggs, capture, 3 pet abilities, server wallet schema	Fun even when nobody opens the shop
Week 2 / P1	Common/Rare/Epic materials, first Common hatch, focus recipe, Gold earn/spend shop	Crafting atomic, first pet reachable in two active matches
Week 3 / P2	Purchasable Gems, premium guaranteed visuals, transaction analytics, low-end mobile QA	Receipt idempotency, transparent prices, fair results
Soft launch	Tune capture rate, player type tiers, Gold/Gem economy, pet visibility	Retention, role engagement, nonpaying participation pass gates

Expansion	6-9 abilities, pet families, events, premium catalog, possible Ranked	Queue, active playerbase and content cadence support

This staged plan supersedes any assumption that a complete nine-ability, multi-rarity economy can be production-ready in one week. The one-week target remains an instrumented competitive vertical slice.

18 Open decisions and formal sign-offs

Decision	Default	Owner must approve?
Launch Gem shop	Yes, purchasable Gem packs and guaranteed premium cosmetics	Yes
Gem payment for Focus Seal	Disabled pending policy + fairness check	Yes / later
Direct egg piece purchase	No	Yes
Gold direct Robux purchase	No	Yes
Gold temporary cosmetic shop	Yes	Yes
Casual pet rarity boosts	0% until A/B shows safe alternative	Yes
Crowned tier at initial launch	No; defer pending acquisition inequality fixes	Yes
Permanent trading	Disabled	Yes
Ranked at initial launch	No, implement optional normalization rules	Yes

Final recommendation. *Gold = earned frequent expression. Gems = purchaseable/earnable permanent premium expression. Eggs = the only primary pet-ability progression resource. Preserve the path from skillful escape or theft to earned pet, rather than selling the whole loop away.*

APPENDIX A Complete multiplayer state machine

Server state	Entry rule	Exit / conflict rule
LOBBY	Player joins or previous match ends	Select eligible participants; do not put inactive late joiners into active reward cohort
SELECT_HUNTER	Prefer volunteers; weighted rotation fallback	Exactly one initial Hunter at regular lobby sizes
CROSSING_1/2/3	Server opens timing and designated safe zone	On escape, atomically upgrade held egg; on valid hit, atomically transfer egg and mark conversion
PHASE_SETTLEMENT	Crossing timer expires / no Runners left	Converted players enter Catcher roster for next crossing; match can end early
MATCH_SETTLEMENT	Three crossings or early infection win	Use stable final snapshot of best earned/stolen tier; issue no more than one primary material
INTERMISSION	Rewards sent / UI displayed	Load shop/incubator; next match begins even if users ignore economy

Authority order: server timestamps determine capture vs safe-zone completion, using a configured reconciliation tolerance. Never have separate LocalScripts award pieces. If a player leaves during settlement, persist entitlements before acknowledging the match result. The same match ID + player ID must settle once; account disconnections should not erase previously committed earnings.

Converted player cooldown continues; no pet or ability equipment swap can be committed mid-match. A Catcher receiving an egg does not become a Runner. Tagged Runners must return to play immediately as Catchers (or at next crossing, depending on the tested anti-snowball config); choose one for initial public build, not both simultaneously.

APPENDIX B Pet architecture and asset plan

Family	Ability	Base/common species	Visual rarity upgrades	Role parity constraint
Dash	Short directional burst	Sand Crab	Otter / Phoenix	Same tested movement range per player tier
Decoy	Moving visual illusion	Mouse	Parrot / Mirror Dragon	Clone must be visible with consistent hit/collision messaging
Scout	Opponent bearing ping	Sparrow	Seagull / Celestial Owl	Works for Runner vs Catcher with proper nearest opponent filtering
Brake	Cancel inertia / pivot	Turtle	Hermit / Time Spirit	Works during pursuit and escape without precision abuse
Spring	Controlled leap	Frog	Hare / Cloud Serpent	No map skips outside reachable geometry
Mist	Short line-of-sight disruption	Puffer	Jelly / Kraken	Enemy telegraph remains clear and mobile friendly
Rebound	Wall bounce	Gecko	Ram / Thunder Beast	No tunneling or clipping across boundary
Drift	Temporary turning control	Minnow	Koi / Leviathan	No invisible speed multiplier
Tracker	Recent enemy footsteps	Firefly	Bat / Ghost Wolf	No wallhack-level advantage in small maps

Each pet config references AbilityFamilyId, PetSpeciesId, RarityId, CosmeticAssetIds, and optional server-selected CasualTierModifier. Pet upgrades cannot independently change more than one ability parameter at a time. Collecting duplicate species yields cosmetic mastery only; do not multiply combat stats.

Render pet privately for owner while in arena, <20% of avatar height; opponents see ability summons only at activation. Give players a local pet-visibility toggle. Never delete or conceal opponent-readable VFX, especially in Ranked. Cosmetics must not obscure collision and hurtboxes.

APPENDIX C Gold / Gem item specification and lifecycle

Item class	Wallet	Can consume repeatedly?	Does it affect winning?	Can buy with Robux?
One-match egg color trail	Gold	Yes	No	No, Gold is earn-only
Lobby celebration burst	Gold	Yes	No	No

Permanent basic emote	Gold	No	No	No
Permanent shell/egg case	Gold	No	No	No
Custom ability VFX color	Gems	No	No; VFX clarity parity required	Yes, via Gem packs
Premium hatch animation	Gems	No	No	Yes
Pet showcase / collection pedestal	Gems	No	No	Yes
Choice of hatch family via Gem Focus Seal	Gems	Each craft	Indirect ability choice; policy review	EXPERIMENT ONLY
Pet material or Legendary instant hatch	None	No	Yes	NOT SOLD

Launch merchant rules: shop shows price in in-game wallet units AND includes a clear Robux-Gem-pack entry, but does not auto-prompt on every insufficient-balance action. Avoid using spend pressure immediately after a capture loss. In-match shop access should not block safe-zone input or pause the round clock unfairly.

Duplicate grants, receipt races, rollback failures, region restrictions and account rejoin during a pending purchase are mandatory QA cases. Client UI is advisory; the authoritative server transaction must resolve products, amounts, wallet debits and item entitlements.

APPENDIX D Focus Seal price-to-effort sensitivity

Potential premium-enabled focus alternative must be compared against the FREE extra-piece option. At Typical simulated acquisition probabilities, one extra exact-tier material takes 2.25/p minutes on average, where p is the probability of that tier per match. Earned-Gem time for a Focus Seal is price / 2.383 Gems-per-match times 2.25 minutes. Amounts below are deliberately comparable, not final prices.

Material tier	One extra piece avg min	Trial Focus Seal (Gems)	Earned-Gem time to seal
Common	2.25	4	3.78 min
Rare	6.25	9	8.50 min
Epic	11.25	16	15.11 min
Crowned FUTURE	56.25	60	56.66 min

This creates a real earn-versus-pay choice without selling the BASE material stack required to hatch. It still allows paid players to accelerate the acquisition of specific abilities, so competitive fairness and paid-random-item policy must be reviewed before activation. If the random recipe reveals only one pet species within a fixed family/tier, the paid route must still disclose any remaining random outcomes and verify per-user restrictions as needed.

APPENDIX E Telemetry event contracts and QA matrix

Event	Key fields / counters	Purpose
match_start	matchId, queueType, playerCount, seed, acquisitionSource	Normalize experiments and player cohort selection
role_selected	volunteered, forced, recentHunterCount	First-Catcher fairness
crossing_complete	phase, tierBefore, tierAfter, abilityFamily, device	Survival and egg transformation validation

capture_confirmed	captorId, victimId, tier, cooldown, collisionReason	Real theft + anti-cheat + progression
piece_awarded	idempotentMatchKey, tier, source, bankedFloor	Anti-duplication and tier creation
gold_source/gold_sink	amount, reason, preBalance, postBalance	Soft economy tracking
gems_source/gems_sink	amount, earnedOrPurchased, reason	Premium ledger integrity
focus_used	tier, extraPieceOrGem, familyId, eligibility	Choice behavior and policy
pet_crafted	speciesId, abilityFamily, rarity, randomFlag	Collection and unlock outcomes
purchase_receipt	productId, transactionId, robux, deliveredGems	Revenue reporting and refunds
return_next_round	postCaptureActivity, firstHatchOwned	Retention feedback

Critical QA: no initial pet; Common first pet earned after two eligible matches; wrong-ability random roll uses expected family pool; focused hatch never picks a different family; family focus debit is extra material exactly once; Crowned disabled grants Epic at Level 4; Catcher steals do not create more than one material for same player; crash-safe transaction replay; mobile ability telegraphs remain readable; paid Gem purchases are granted once and can be refunded/reconciled.

Security QA: injected client capture packets; out-of-range tags; forged high-tier eggs; match reward replays; repeated purchase receipts; concurrent transactions from two servers; manual clock manipulation; AFK farming and teaming between alt accounts. All wallet writes are server-side.

APPENDIX F Source reliability and model limitations

Roblox Creator Hub documentation is authoritative for platform behavior and monetization policy. Huss Valley gamepasses are from public third-party Rolimon listings; full developer-product inventories and revenue are not known. Informal public Huss Valley guides are secondary context; never claim access to its server settings, Roblox financial dashboard, conversion rates, acquisition mix, or retention data.

The model does not jointly simulate player skill, real arena geometry, latency, paid ability bonuses, team compositions and repeated interactions. It samples categorical material outcomes and a separate credited-theft indicator to stress-test economy units. This is NOT a causal outcome model. Before launch, fit reward distributions to actual match telemetry and replace all nominal source/sink prices with tested values.

SOURCES AND METHODS

[S1] Roblox Discovery: https://create.roblox.com/docs/discovery

[S2] Roblox Discovery FAQ: https://create.roblox.com/docs/discovery-faq

[S3] Roblox Monetization Foundations: https://create.roblox.com/docs/production/game-design/monetization-foundations

[S4] Roblox Balance Virtual Economies: https://create.roblox.com/docs/production/game-design/balance-virtual-economies

[S5] Roblox Monetization Analytics: https://create.roblox.com/docs/production/analytics/monetization

[S6] Huss Valley Rolimon public passes: https://www.rolimons.com/game/107535308163741

[S7] Huss Valley guide: gems/coins (secondary): https://robloxguide.wiki/huss-valley/guides/huss-valley-gems-coins

[S8] Roblox Paid Random Items: https://create.roblox.com/docs/production/monetization/paid-random-items

[S9] Roblox Developer Products: https://create.roblox.com/docs/production/monetization/developer-products

[S10] Ryan/Rigby/Przybylski self-determination video game paper: https://selfdeterminationtheory.org/SDT/documents/2006_RyanRigbyPrzybylski_MandE.pdf

[S11] Nunes/Dreze Endowed Progress Effect: https://academic.oup.com/jcr/article-abstract/32/4/504/1787425

[S12] Freeman et al competitive purchases and fairness: https://researchwith.njit.edu/en/publications/pay-to-win-or-pay-to-cheat-how-players-of-competitive-online-game/

[S13] Roblox Monetization and 70% creator share: https://create.roblox.com/docs/get-started/monetization

[S14] Roblox DevEx rates: https://create.roblox.com/docs/production/monetization/developer-exchange

[S15] Third party crate probability report (not official): https://hussvalleyguide.com/calculator/

[S16] Roblox Price Optimization: https://create.roblox.com/docs/production/monetization/price-optimization

[S17] Roblox Analytics Events: https://create.roblox.com/docs/production/analytics/event-types

[S18] Roblox Contextual Purchases: https://create.roblox.com/docs/production/game-design/contextual-purchases

Mathematical methods: categorical per-match reward-outcome sampling (150,000 observations per cohort), binomial probability for qualifying eggs, geometric/negative-binomial expected acquisition duration, coupon collector expectation H_K, and constructed Robux-to-USD scenarios. Complete reproducible model code and CSV summaries are separate working artifacts. All experiment thresholds and hypothetical player tier distributions are unvalidated design inputs.